MIL-F-0010117B(MU)
14 August 1969
USED IN LIEU OF
MIL-F-10117A
4 February 1953

MILITARY SPECIFICATION

FACEPIECE, COMBAT, M9

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-F-10117A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the M9A1 Protective Mask facepiece.

1.2 Classification.

1.2.1 Size. Facepieces shall be of the following sizes, as specified (see 6.2):

Small (E5-2-764 or E5-2-804 faceblank)
Medium (E5-2-745 or E5-2-805 faceblank)
Large (E5-2-821 faceblank)

1.2.2 Canister mounting. Facepieces shall have the canister mounting on the right or left cheek, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 4240

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SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel, Strapping, Flat.
TT-W-572 - Wood Preservative, Water-Repellent.
UU-C-282 - Chipboard.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-F-10135 - Faceblanks, Field Protective Mask.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

D5-2-758 - Facepiece, Combat, M9.
D5-22-47 - Faceform, Gas Mask, M5 (for Large and Medium Facepiece).
D5-22-52 - Faceform, Gas Mask, M5 (for Small Facepiece).
E38-6-327 - Q106 Inlet and Outlet Valve Resistance Indicator.
C136-10-87 - Nosecup to Faceblank Adhesion Tester.
DL136-42-750 - M14 Gas Mask Leakage Tester.
DL136-45-202 - M4A1 (Mod.) Outlet Valve Leakage Indicator.
D150-2-1 - Marking Diagram for Shipping Containers.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of this publication should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to Drawing D5-2-758 and the subsidiary drawings listed thereon.

3.2 Manufacture and assembly. The facepiece shall be manufactured and assembled in accordance with Drawing E5-2-759 and subsidiary drawings listed thereon.

3.3 Outlet valve assembly resistance. The outlet valve assembly shall have an airflow resistance not exceeding a pressure differential of 22 millimeters of water at an airflow rate of 85 liters per minute when tested as specified in 4.4.4.1.

3.4 Outlet valve assembly leakage. The outlet valve assembly shall not leak when tested as specified in 4.4.4.2.

3.5 Facepiece assembly leakage. Leakage of the facepiece assembly (excluding leakage at the periphery) shall not exceed 0.003 percent di-octylphthate (DOP) smoke penetration when tested as specified in 4.4.4.3.

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3.6 Nosecup to faceblank adhesion. Nosecup to faceblank adhesion, before and after aging, shall be not less than 5 pounds when tested as specified in 4.4.4.4. Separation of either side of the nosecup from the faceblank shall not exceed 1/2 inch below the curled edge of the nosecup.

3.7 Embrittlement. Sections (4.3.3.1) of the nosecup-to-faceblank bonded area shall show no evidence of cracking when tested as specified in 4.3.3.2.

3.8 Preproduction. Prior to the start of regular production a preproduction sample of facepieces shall be produced, in accordance with this specification, for examination and tests (4.3). In addition to the requirements of this specification the preproduction sample shall pass the tensile strength, ultimate elongation, and tensile stress requirements of MIL-F-10135.

3.9 Workmanship. The facepiece shall be free from contamination (grease, oil, or foreign matter) and damage such as permanent set of rubber components, scratched or distorted lenses, cracks, dents, cuts, tears, and malfunctioning of hardware.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 25 facepieces shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The preproduction sample shall be examined for all the requirements of the drawings and of this specification.

4.3.2.2 For destructive tests. The preproduction sample shall be tested in accordance with 4.3.3.

4.3.3 Tests.

4.3.3.1 Specimens. Specimens shall be taken and tested as follows: Take 20 facepieces from the preproduction sample, at random. Condition the facepieces a minimum of 24 hours at 75° ± 5° F. Remove the forehead section from each faceblank by cutting along a line just above the temple harness yokes and eyepiece openings. Cut the remainder of the facepiece along the parting line. Die out two dumbbell specimens from each forehead section and use for unaged tests. Die out one specimen from the cemented area and another from the adjacent uncemented area of the aged sections used for embrittlement testing (4.3.3.2). Take both specimens from the faceblank side opposite the canister stem. Carefully remove and discard the piece of nosecup attached to the faceblank specimen. Buff the cemented specimen as specified in method 1111 of FED-STD-601 to a depth not greater than necessary to remove all visual evidence of cement or gouging which may have resulted from detaching the nosecup. Test the unaged (forehead) specimens and the aged (cheek) specimens for tensile strength, ultimate elongation, and tensile stress as specified in MIL-F-10135. Average the results obtained on unaged specimens, on aged specimens taken from the cemented area, and on aged specimens taken from uncemented areas. Discard all results not within 10 percent of the average. Then group the results of all the aged specimens and determine the average of the aged and unaged specimens separately.

4.3.3.2 Embrittlement. Embrittlement shall be determined as follows: Flex the bonded area of each cemented section between the fingers by turning the chin portion outwardly, folding it over on itself (180° bend), and rolling several times between the fingers. Repeat this procedure after the cemented sections have been aged in a circulating air oven for 7 days at 158° ± 2° F and conditioned at 75° ± 5° F for at least 24 hours.

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4.3.3.3 Nosecup to faceblank adhesion. The aged cemented sections tested for embrittlement in 4.3.3.2 shall also be tested for nosecup to faceblank adhesion as specified in 4.4.4.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the facepieces, of one size, produced by one manufacturer from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of faceblanks, nosecups, or outlet valves shall be represented in any one lot of finished facepieces.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, using level S-3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample facepieces shall be examined and tested in accordance with the classification of defects and MIL-STD-105. Each facepiece shall be tested for critical defects.

4.4.3.2 For destructive tests. Sample facepieces shall be tested in accordance with 4.4.4.4. The acceptance number is 4.

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4.4.3.3 Classification of defects

(a) Facepiece, combat, M9 - assembly (Dwg E5-2-759).

Categories	Defects	Acceptance standards
Critical:		
1	Outlet valve assembly leakage	4.4.4.2
2	Facepiece assembly leakage	4.4.4.3
Major:	AQL 0.25 percent defective	
101	Outlet valve assembly resistance	4.4.4.1
	AQL 1.0 percent defective	
102	Component missing, incorrect, damaged, or incorrectly assembled	
103	Canister mounting, thread nonfunctioning	
104	Hardware inoperative	
105	Permanent set	
106	Incorrect riveting of temple tabs	
107	Sharp edges on metal components	
108	Lot numbering marking incorrect	
109	Workmanship (3.9)	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing or incorrect	
202	Taping inadequate	
203	Marking missing or illegible (other than 108)	

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4.4.4 Tests.

4.4.4.1 Outlet valve assembly resistance. Exhalation resistance of the outlet valve assembly shall be determined with the Q106 Inlet and Outlet Valve Resistance Indicator (E38-6-327).

4.4.4.2 Outlet Valve Assembly Leakage. Leakage of the outlet valve assembly shall be determined with the M4A1 (Mod.) Outlet Valve Leakage Indicator (DL136-45-202).

4.4.4.3 Facepiece assembly leakage. Leakage of the facepiece assembly shall be determined with the M14 Gas Mask Leakage Tester (DL136-42-750).

4.4.4.4 Nosecup to faceblank adhesion. Adhesion of the nosecup to the faceblank shall be determined with the Nosecup to Faceblank Adhesion Tester (C136-10-87). Repeat this test on the aged sections used in embrittlement testing of preproduction.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A.

5.1.1 Unit-packaging. The contour of the facepieces shall be protected by inserting a faceform as specified on Drawing D5-22-47 or D5-22-52 as applicable. The chinpiece of the faceform shall rest in the interior chinportion of the facepiece and the brow piece support the brow portion. The head harness shall be neatly folded against the back of the faceform and the facepiece tied in a compact package.

5.1.2 Intermediate packaging. Two facepieces, of one type and size packaged as specified in 5.1.1, shall be packaged in a style F.O.L. fiberboard box conforming to W6c of PPP-B-636. The facepieces shall be separated by either a type CF, class domestic, variety SW fiberboard spacer conforming to PPP-F-320 or a 1/16 inch plain chipboard spacer conforming to UU-C-282. The fiberboard box shall be closed and sealed in accordance with the appendix thereto.

5.2 Packing, level A. Ten intermediate packages, packaged as specified in 5.1.2, shall be packed in a style 2, 2-1/2, or 4 nailed wood box conforming to PPP-B-621. The box shall be fitted with a waterproof case liner conforming to MIL-L-10547 and folded and sealed in accordance with the appendix thereto. The wooden box shall be closed and strapped in accordance with the appendix thereto. Galvanized strapping conforming to type I, class B, of QQ-S-781 shall be used. The boxes or the finished wood parts thereof shall be completely immersed for a minimum of three minutes in a water-repellent wood preservative conforming to type II, composition A of TT-W-572.

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5.2.1 Packing, level B. Ten intermediate packages shall be packed the same as that specified in 5.2, except the shipping container shall be a style RSC fiberboard box conforming to V3c of PPP-B-636. The box shall be closed and sealed in accordance with the appendix thereto.

5.2.2 Packing, level C. The facepieces shall be packed as specified in 5.2.1 except the shipping container shall be a style RSC fiberboard box conforming to Type CF, class domestic, variety SW, grade 200, of PPP-B-636. The box shall be closed and sealed in accordance with the appendix to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and intermediate packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with D150-2-1. All markings of the shipping container shall be clear, legible, and permanent after application of the wood preservative specified in 5.2.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. The facepiece covered by this specification is intended for use in masks with a canister mounting on the right or left cheek.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of facepiece and location of canister mounting.
(c) Level of packing.
(d) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation before award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not sample is acceptable.

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6.3 Component specifications.

Title	Number
C3 Eyering	MIL-E-51274
C8R1 Headharness	MIL-H-10120
C15 Outlet Valve	MIL-V-10134
Canister Mounting	MIL-M-10156
Inlet Valve Discs	MIL-D-10131
Faceblanks Field Protective Mask	MIL-F-10135
Lenses, Ophthalmic, Simple	MIL-L-50064
Nosecups, Field Protective Mask	MIL-N-10130
Nosecup Inlet Valve Disc	MIL-D-10115

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A390

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents, or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

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